

Mathematical formulation of the external Meta-TN extension

The external Meta-TN extension was used to provide an independently estimated tumor-normal biological-support component when StaBiCut was applied to the tumor-only GSE39582 survival cohort. It extends, rather than identically re-estimates, the tumor-normal component used in the TCGA discovery analysis. The discovery component recorded directional agreement as a binary score, whereas the external Meta-TN component provides a graded score between 0 and 1 based on the reproducibility, magnitude, and meta-analytic support of tumor-normal expression differences across independent cohorts. The Meta-TN component occupied the original tumor-normal weight of 0.15 in the five-component StaBiCut score. Its addition did not change the GSE39582 expression preprocessing, cutoff scan, selected cutoff, bootstrap resampling, cutoff-plausibility score, bootstrap hazard-direction consistency, or group-balance score. It affected only the final external composite score and the resulting gene ranking.

Cohort-level preprocessing and sample definition

Four independent colorectal cancer datasets containing primary tumor and normal colorectal mucosa were analysed: GSE21510, GSE32323, GSE41258, and GSE68468. Each dataset was processed separately, and expression matrices from different studies or Affymetrix platforms were not pooled. Probe identifiers were mapped to gene symbols using platform-specific annotation resources (GPL570: *hgu133plus2.db*; GPL96: *hgu133a.db*). When multiple probes mapped to the same gene, the probe with the highest median expression across all samples in that cohort was retained.

For each dataset, predefined terms in the GEO sample annotations were used to classify samples as primary colorectal tumor, normal colorectal mucosa, or excluded. Samples matching an exclusion term were removed first. Samples matching only tumor terms or only normal terms were assigned accordingly.

When both sets of terms were present, an explicit “normal” annotation took priority; otherwise, the sample was classified as tumor. All sample assignments and the annotation text used for classification were exported for manual review. Tumor and normal samples were analysed as two groups. Even when a dataset contained matched specimens from the same patient, the implemented calculation did not use within-patient expression differences.

Cohort-specific standardized tumor–normal effects

For gene j in cohort k , let $\bar{x}_{T,jk}$ and $\bar{x}_{N,jk}$ denote the mean expression values in tumor and normal samples, respectively; $n_{T,jk}$ and $n_{N,jk}$ their sample sizes; and $s_{T,jk}$ and $s_{N,jk}$ their within-group standard deviations. The pooled within-cohort standard deviation was

$$s_{p,jk} = \sqrt{\frac{(n_{T,jk} - 1)s_{T,jk}^2 + (n_{N,jk} - 1)s_{N,jk}^2}{n_{T,jk} + n_{N,jk} - 2}}.$$

The uncorrected standardized mean difference was

$$d_{jk} = \frac{\bar{x}_{T,jk} - \bar{x}_{N,jk}}{s_{p,jk}}.$$

The small-sample correction was applied:

$$J_{jk} = 1 - \frac{3}{4(n_{T,jk} + n_{N,jk}) - 9},$$

and cohort-specific Hedges' g was

$$g_{jk} = J_{jk}d_{jk}.$$

Its sampling variance and standard error were estimated as

$$\begin{aligned} \text{Var}(g_{jk}) &= \frac{n_{T,jk} + n_{N,jk}}{n_{T,jk}n_{N,jk}} + \frac{g_{jk}^2}{2(n_{T,jk} + n_{N,jk} - 2)}, \\ SE(g_{jk}) &= \sqrt{\text{Var}(g_{jk})}. \end{aligned}$$

Positive g_{jk} values indicate higher expression in tumor than in normal mucosa, whereas negative values indicate lower expression in tumor. A cohort was included in the gene-level meta-analysis only when both g_{jk} and $SE(g_{jk})$ were finite and $SE(g_{jk}) > 0$.

Cross-cohort meta-analysis and directional consistency

For each gene j , valid cohort-specific Hedges' g estimates were combined using a random-effects meta-analysis fitted by restricted maximum likelihood when at least two valid cohort estimates were available. Let $\hat{g}_j$ denote the pooled Hedges' g , $SE(\hat{g}_j)$ its standard error, and p_j the corresponding two-sided meta-analytic P value. The 95% confidence interval, between-cohort variance τ_j^2 , and I_j^2 statistic were retained for reporting.

A fixed-effect inverse-variance calculation was included only as a software fallback if random-effects fitting was unavailable. Random-effects meta-analysis was the intended primary estimator. The direction of the cohort-specific effects was also summarized. Let $K_{-,j}$ and $K_{+,j}$ denote the numbers of valid cohorts with $g_{jk} < 0$ and $g_{jk} > 0$, respectively, and let K_j denote the total number of valid cohort effects. The majority tumor-normal direction D_j and the proportion of cohorts supporting the majority direction C_j were defined as

$$D_j = \begin{cases} \text{Tumor_down,} & K_{-,j} > K_{+,j}, \\ \text{Tumor_up,} & K_{+,j} > K_{-,j}, \\ \text{Mixed_or_tie,} & K_{-,j} = K_{+,j}, \end{cases}$$

$$C_j = \frac{\max(K_{-,j}, K_{+,j})}{K_j}.$$

Effects equal to zero contributed to K_j but to neither directional count.

Construction of the Meta-TN support score

The pooled result was converted to a bounded prioritization score using three quantities: the proportion of cohorts supporting the majority direction, the magnitude of the pooled standardized effect, and a P -value-derived measure of meta-analytic support.

The effect-magnitude component E_j was

$$E_j = \min\left(\frac{|\hat{g}_j|}{2}, 1\right),$$

and P -value-derived component P_j was defined as

$$P_j = \min\left(\frac{-\log_{10}[\max(p_j, \varepsilon)]}{10}, 1\right),$$

where ε denotes the smallest positive numerical value used to avoid evaluating $\log_{10}(0)$. Before applying the prespecified tumor-normal direction requirement, the base Meta-TN support score was defined as

$$S_j^{base} = C_j E_j P_j^*, \quad 0 \leq S_j^{base} \leq 1.$$

The multiplicative form was used so that a high score required concurrent support from cross-cohort direction agreement, pooled effect magnitude, and the P -value-derived meta-analytic support component. The constants 2 and 10 were fixed scaling parameters used to place the magnitude and P -value-derived terms on bounded 0-1 ranges. Thus, $|\hat{g}_j| \geq 2$ give the maximum magnitude contribution, and values with $p_j \leq 10^{-10}$ receive the maximum P -value contribution. These constants were not estimated from GSE39582 or optimized against the external ranking, and their saturation points should not be interpreted as biological, clinical, or statistical decision thresholds. Because the meta-analytic P value depends on both effect magnitude and estimation precision, the magnitude and P -value-derived terms were not interpreted as independent evidence; the latter was used only as a bounded measure of meta-analytic support within the prioritization score.

Concordance with the prespecified tumor-normal direction

For the present 11-gene panel, the expected tumor-normal direction was defined from the same longitudinal multi-omics evidence used to establish the gene-specific StaBiCut direction priors. Accordingly, genes assigned as protective_high in this study were expected to be lower in tumor than in normal mucosa, whereas genes assigned as adverse_high were expected to be higher in tumor. This mapping was specific to the direction-prior scheme used for the present candidate panel and should not be interpreted as a general equivalence between prognostic direction and tumor-normal expression change. Let $I_j^{prior} = 1$ when D_j matched this expected tumor-normal direction and $I_j^{prior} = 0$

otherwise:

$$I_j^{prior} = \begin{cases} 1, & D_j = D_j^{expected}, \\ 0, & otherwise, \end{cases}$$

The prior-concordant support score was

$$S_j^{prior} = I_j^{prior} S_j^{base}.$$

For integration with GSE39582, a final consistency check compared the sign of the pooled Hedges' g with the high-versus-low hazard direction observed at the selected external cutoff. Let HR_j^{ext} denote the hazard ratio for the high-expression group relative to the low-expression group, and define

$$I_j^{HR} = I\{\text{sign}(\hat{g}_j)\text{sign}[\log(HR_j^{ext})] = 1\}.$$

The Meta-TN component entering the extended composite score was

$$S_j^{MetaTN} = I_j^{HR} S_j^{prior}.$$

Accordingly, a positive pooled Hedges' g was required to accompany an adverse high-expression association in GSE39582, whereas a negative pooled Hedges' g was required to accompany a protective high-expression association. Otherwise, the Meta-TN component was set to zero. Because the GSE39582 cutoff scan had already been constrained by the same gene-specific hazard-direction priors, this step served only as a final consistency check and did not alter the selected cutoff or the external survival model.

Tumor-only and Meta-TN-extended external composite scores

Let $S_{boot,j}$, $S_{density,j}$, $S_{hazard,j}$, and $S_{balance,j}$ denote the GSE39582 bootstrap-stability, cutoff-plausibility, bootstrap hazard-direction consistency, and group-balance scores, respectively. When the tumor-normal component was unavailable, the tumor-only external composite score was normalized by the sum of the retained original weights:

$$S_j^{no\ TN} = \frac{0.40S_{boot,j} + 0.20S_{density,j} + 0.15S_{hazard,j} + 0.10S_{balance,j}}{0.85}.$$

The tumor-only score and its corresponding rank were calculated in the upstream GSE39582 StaBiCut runner. The Meta-TN extension read these

previously generated values and did not reconstruct or alter the four GSE39582-derived component scores. After incorporation of the external Meta-TN component, the original five-component weight structure was reinstated:

$$S_j^{ext+MetaTN} = 0.40S_{boot,j} + 0.20S_{density,j} + 0.15S_{hazard,j} + 0.10S_{balance,j} + 0.15S_j^{MetaTN}.$$

Genes were ranked in descending order of the corresponding composite score; genes with identical scores received the same rank. Addition of the external Meta-TN component affected only the final Meta-TN-extended composite score and ranking. The selected GSE39582 cutoff and all four GSE39582-derived component scores remained unchanged.